EXP:09

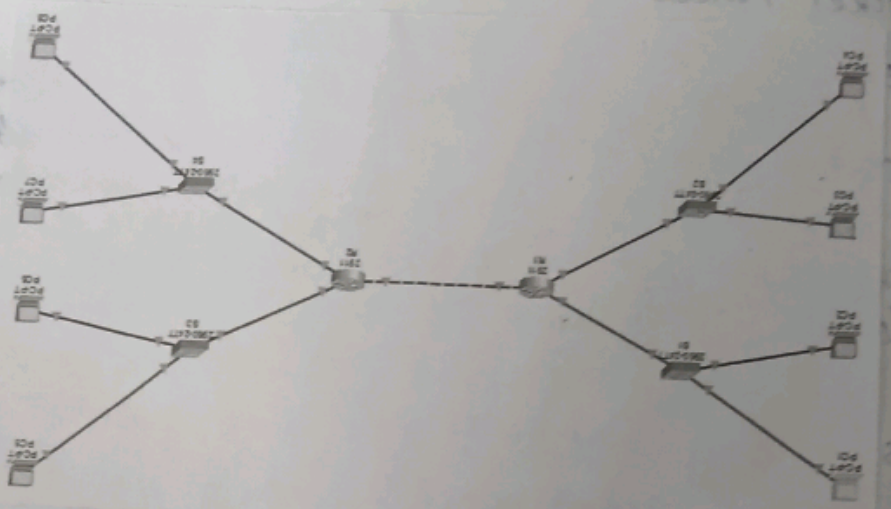
IMPLEMENTATION OF SUBNETTING USING THE CISCO PACKET TRACER

Aim:

To implement classless IP subnetting and configure communication between different subnets using cisco packet tracer.

STEPS:

1. Subnetting is the process of dividing a large IP network into smaller networks called subnets
2. classless subnetting uses a variable-length subnet mask instead of relaying only on the default class based-masks.
3. In this experiment, the network 192.16.8.1.0/24 is dividing using the /28 subnet mask.
4. The subnet mask for /28 is 255.255.255.224
5. A /28 subnet provides 32 total addresses and 30 usuable host addresses
6. subnetting helps in efficient IP address utilization and reduces broadcast traffic.
7. Different subnets can communicate through a router.



Result: Thus a classless IP subnetting was successfully implemented in cisco Packet tracer and connectivity between the different subnets was verified using the ping command.